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Investigation of transient transport dynamics induced by compact torus injection in the EAST tokamak

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Abstract

The rapid increase in plasma core electron temperature triggered by cold plasma injection has long been recognized as compelling evidence of nonlocal transport. This study reports on experiments investigating ‘nonlocal’ thermal transport via compact torus (CT) injection in the plasmas of the EAST tokamak. In low-density plasmas, the core temperature increase following CT injection is accompanied by a drop at the edge. Owing to the CT’s high density, high velocity, and self-organization, rapid penetration into deeper plasma regions was experimentally achieved. In this experiment, the evolution of plasma electron density during CT injection was measured using a high-time-resolution reflectometry for density profile measurement. The evolution of the electron density profile following CT injection is characterized by a rapid increase in core density and subsequent quick flattening of the profile. It has also been found that turbulence plays a crucial role in the mediation of ‘nonlocal’ thermal transport phenomena. The intensity of the quasi-coherent modes (QCMs) in core plasmas, identified by the analysis of Doppler backscattering signals, decreased following CT injection. The simulations based on the trapped gyro-Landau fluid transport model demonstrate that the QCM might be associated with the trapped electron mode (TEM). The observed increase in plasma core temperature after CT

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injection may be attributed to the stabilization of TEM due to reduction in the core electron density gradient. The simulation results agree well with the experimentally measured core temperature increase after CT injection.

Keywords: transient thermal transport, CT injection, turbulence, tokamak

(Some figures may appear in colour only in the online journal)

1. Introduction

Understanding the transport mechanisms in magnetically confined plasmas is critical for advancing nuclear fusion research, and mastering the dynamics of transient plasmas is equally indispensable for controlling future fusion reactors [1]. Experiments conducted in the TEXT tokamak in 1995 demonstrated that adding a perturbed cold pulse at the plasma boundary with low collisionality led to a decrease in the boundary temperature while the core temperature rose rapidly [2]. Subsequently, this phenomenon was reported in multiple tokamaks and stellarators (TFTR [3], Tore Supra [4], ASDEX-U [5], JET [6], HL-2A [7], EAST [8], Alcator C-Mod [9], KSTAR [10], J-TEXT [11], LHD [12] and RTP [13]). The decrease in the boundary temperature was caused by the energy exchange resulting from the injection of cold neutral particles. However, the mechanism of the rapid increase in the core temperature (much faster than the energy confinement time) following cold perturbation pulse in the boundary remains poorly understood.

Early studies developed empirical, marginal stability and self-organized criticality models for nonlocal transport. Fractional diffusion operator-based nonlocal models [14] were later used to explain experiments on JET. The turbulence spreading [15] and residual stress models were also proposed to reconstruct internal rotation reversal and the polarity reversal of cold pulses. Furthermore, measurements conducted on EAST following supersonic molecular beam injection (SMBI) have revealed results concerning plasma electron temperature, density, and density fluctuations during the nonlocal transport phase. Combining experimental results from the Alcator C-Mod device with modeling studies based on trapped gyro-Landau fluid (TGLF), the core temperature reversal phenomenon [16] was well explained by the stabilization of turbulence associated with the trapped electron mode (TEM), resulting from the rapid flattening of the electron density profile induced by the injection of ablated impurities. However, in Alcator C-Mod, the evolution of the electron density profile was not experimentally measured with sufficiently high temporal resolution. The electron density profile distribution used in the simulation was constructed as a skewed Gaussian distribution profile with an inward-propagating characteristics. Subsequent identical impurity injection experiments on DIII-D [17], which combined density evolution data from high temporal resolution reflectometry with the model, successfully recovered the temperature reversal phenomenon. In ASDEX Upgrade [18], TGLF simulations were conducted to explain the results from laser blow-off impurity injection experiments under two heating conditions. A rapid increase

in the central electron temperature occurred exclusively with electron-dominated heating, whereas such a rapid increase was not found with ion-dominated heating.

Two key issues arise from the temperature reversal following cold pulse injection: (1) the increase of the core electron temperature and (2) the rapidity of the change in core electron temperature. The core electron temperature rise has been explained by the competition between TEM and ion temperature gradient (ITG) modes, but the mechanism underlying the extremely rapid change of core electron temperature remains poorly validated. Turbulence was observed to play a crucial role in nonlocal transport triggered by SMBI in EAST. However, the SMBI penetration depth was only a few centimeters into the separatrix, and the response time of the core electron density was approximately 10 ms after SMBI. In addition, it was found that the change in core electron density occurred before the change in the core turbulence [8]. It should be noted that compact torus (CT), as a method of central fueling [19–23], can be rapidly injected into the core plasmas much deeper than SMBI since CT is characterized by its high velocity and density, and its stable magnetic field configuration. Building on existing experiments and incorporating the rapid core penetration achieved by CT injection, investigations into ‘nonlocal’ thermal transport phenomena are expected to further elucidate the mechanisms underlying ultrafast core temperature reversal.

The remainder of this paper is organized as follows. Section 2 describes the experimental setup for CT injection experiments on EAST, including the CT injection system configuration, heating methods, diagnostic arrangements, and relevant plasma parameters. The primary experimental results are presented in section 3, encompassing electron temperature responses to CT injection under high and low plasma density scenarios, and the evolution of density and temperature profiles, and linear stability analysis based on the TGLF model. Section 4 summarizes the results and discusses future research directions.

2. Experimental setup

The EAST tokamak is a fully superconducting device featuring a noncircular poloidal cross-section [24]. It has a major radius $R = 1.9$ m, a minor radius $a = 0.45$ m, maximum magnetic field of 3.5 T, plasma current of 1 MA, flexible double-null or single-null plasma geometry, and an ITER-like configuration [25]. The experimental results reported in this paper were obtained from a series of plasma discharge experiments on the EAST tokamak. These experiments aimed to

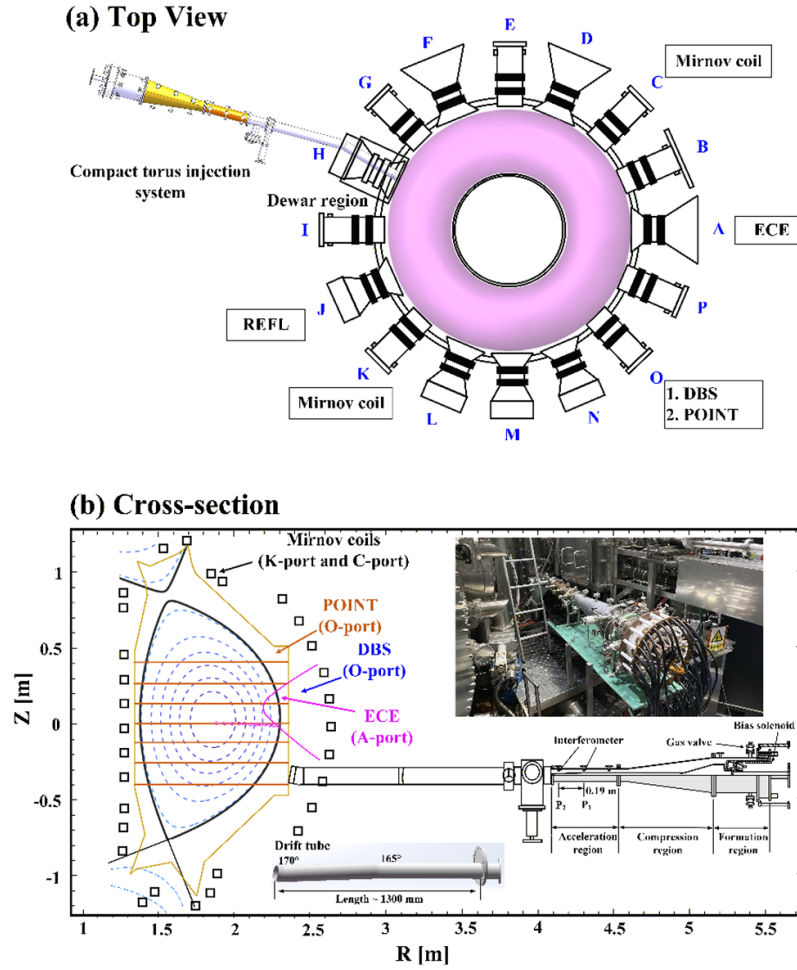


Figure 1. (a) The schematic top view of CT injection system and essential diagnostics on EAST, such as ECE, DBS, REFL and POINT. (b) The cross-section of CT injection system on EAST. The pink hollow circles indicate the projection view of ECE, the blue dash line is the trajectory of the DBS ray, the orange horizontal line is the measurement position of POINT with their toroidal locations shown in (a).

investigate the rapid response of the plasma to cold pulses through fast CT injection under various background density levels. These experiments were conducted under the following parameters: plasma current $I_p = 400$ kA, toroidal magnetic field $B_T = 2.5$ T, a safety factor at the 95% of the toroidal flux $q_{95} \approx 7$. The plasma was heated by means of ECRH.

Figure 1(a) shows the top view of the CT injection system during the experimental campaign. The CT system was installed on the shared H port of EAST and was connected to a 100 mm flange opening positioned 195 mm below the midplane [19]. Figure 1(b) shows a cross-section view of the CT injection system and the essential diagnostics. Due to space constraints, a 1300 mm drift tube was installed in the Dewar region to guide the CT plasma toward the first wall on the low-field side. To investigate the impact of CT injection on ‘nonlocal’ transport processes, high-time-resolution diagnostics were employed to measure the plasma evolution following CT injection. The electron temperature was measured by the electron cyclotron emission radiometer (ECE) [26]. The evolution process of the density after CT injection can be measured using the reflectometers (REFL) [27] and the polarimeter interferometer (POINT) [28]. Additionally, a

multi-channel Doppler backscattering (DBS) system [29–31] measures the electron density fluctuation and perpendicular velocity.

3. Experimental results

3.1. Characteristics of the CT injection system on EAST

The main purpose of CT injection into EAST is to demonstrate central fueling under high magnetic field conditions. Central fueling can enhance the tritium burn-up rate, which is a key enabler for reducing the requirements for tritium breeding rate and tritium handling capability in future fusion reactors. The EAST CT system was operated at a 2 Hz repetition rate [32] and features a wide injection parameter window [33]. The typical waveforms of the CT plasma parameters with a 2.4 T background toroidal magnetic field in EAST are shown in figure 2(a). The entire formation and acceleration current waveforms lasted approximately 100 μ s and only the first half cycle (< 20 μ s) in each discharge was responsible for CT formation and acceleration. Fiber-optical laser interferometers measured line-averaged density waveforms

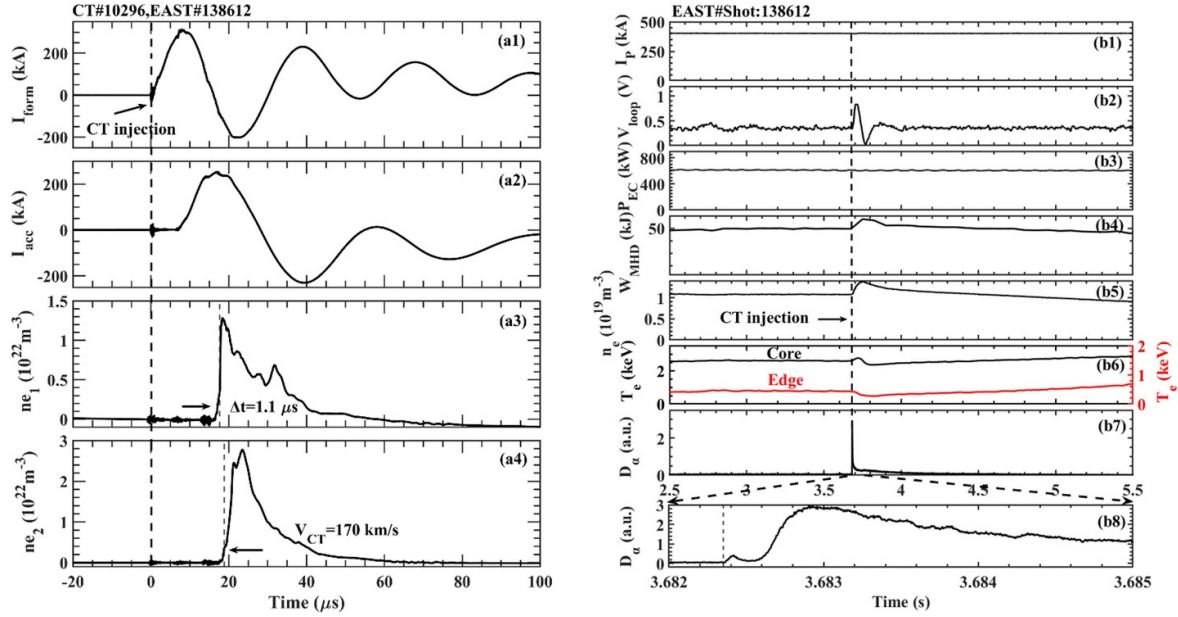


Figure 2. (a) EAST-CTI discharge parameters: (a1), (a2) formation current and acceleration discharge current of CT, (a3), (a4) electron density measured at positions P_1 and P_2 . (b) Temporal evolution of EAST plasma parameters (shot: 138612) around CT injection time: (b1) plasma current, (b2) loop voltage, (b3) power of ECRH, (b4) plasma stored energy, (b5) central-line-averaged electron density, (b6) electron temperature, (b7), (b8) D_α emission intensity.

of CT at two locations (P_1 and P_2), separated by 0.19 m as sketched in figure 1(b), as CT was accelerated through the coaxial accelerator. The peak value of the line-averaged plasma density up to $2.1 \times 10^{22} \text{ m}^{-3}$ was measured. The CT velocity of approximately 170 km s^{-1} was deduced based on time-of-flight of the two density waveforms. Combined with the estimated CT velocity the particle number at CT ejection was about 1.9×10^{20} (corresponding to a mass of 0.64 mg), with a corresponding kinetic energy density of approximately $1 \times 10^6 \text{ J m}^{-3}$ or 8.7 kJ in total. It is precisely due to its high directional kinetic energy that the CT can penetrate deeply into the magnetic field and plasma and rapidly deposit its particle into the core region of the plasma, possibly through a magnetic reconnection process [34, 35].

Figure 2(b) shows the waveforms of the EAST parameters shortly before and after CT injection. It takes approximately 200–400 μs for the CT to travel more than 1.3 m through the drift tube to the plasma boundary. The vertical dashed lines indicate the time of CT injection. After CT injection, the stored energy stored increased by 8 kJ, corresponding to an increase of approximately 16%. This change can be attributed to the energy injected by the CT system itself. Concurrently, the line-averaged central density rose by roughly 30% following the injection. In addition to these changes, a rapid enhancement of the D_α signal was observed. The edge electron temperature decreased after the CT injection. However, the core electron temperature increased momentarily while the edge temperature was decreasing.

3.2. Transient thermal transport characteristics of plasma in EAST triggered by CT injection

During the CT injection experiments, the EAST tokamak was operated under the double-null configuration. The line-averaged electron density was measured using the POINT interferometer, and the time-resolved electron temperature was measured using an ECE system. The plasma was heated by ECRH, and the plasma current was maintained at $I_p = 400 \text{ kA}$. Figure 3 shows the expanded waveforms of the electron temperatures measured by ECE at different radial locations in the region $0.3 < \rho < 0.6$ (where ρ is the normalized minor radius) following CT injection under three different levels of the line-averaged electron density ($1.2 \times 10^{19} \text{ m}^{-3}$, $1.4 \times 10^{19} \text{ m}^{-3}$, $2.0 \times 10^{19} \text{ m}^{-3}$) measured by POINT in EAST. After CT injection, the measured electron temperature in EAST showed different features at different radial locations. The temperature responses to CT injection also depended on the baseline density levels. As shown in figures 3(a) and (b), the temperatures of the inner channels increased for the two lower density levels, whereas figure 3(c) shows a decrease in electron temperature at all radial locations under high-density conditions. At outer radial locations, the temperatures decreased for all three baseline densities. It should be noted that as the density increases, the rate of temperature increase progressively diminished from baseline density of $1.2 \times 10^{19} \text{ m}^{-3}$ to $1.4 \times 10^{19} \text{ m}^{-3}$ and ultimately decreases when the baseline density is $2.0 \times 10^{19} \text{ m}^{-3}$, proving that the temperature reversal

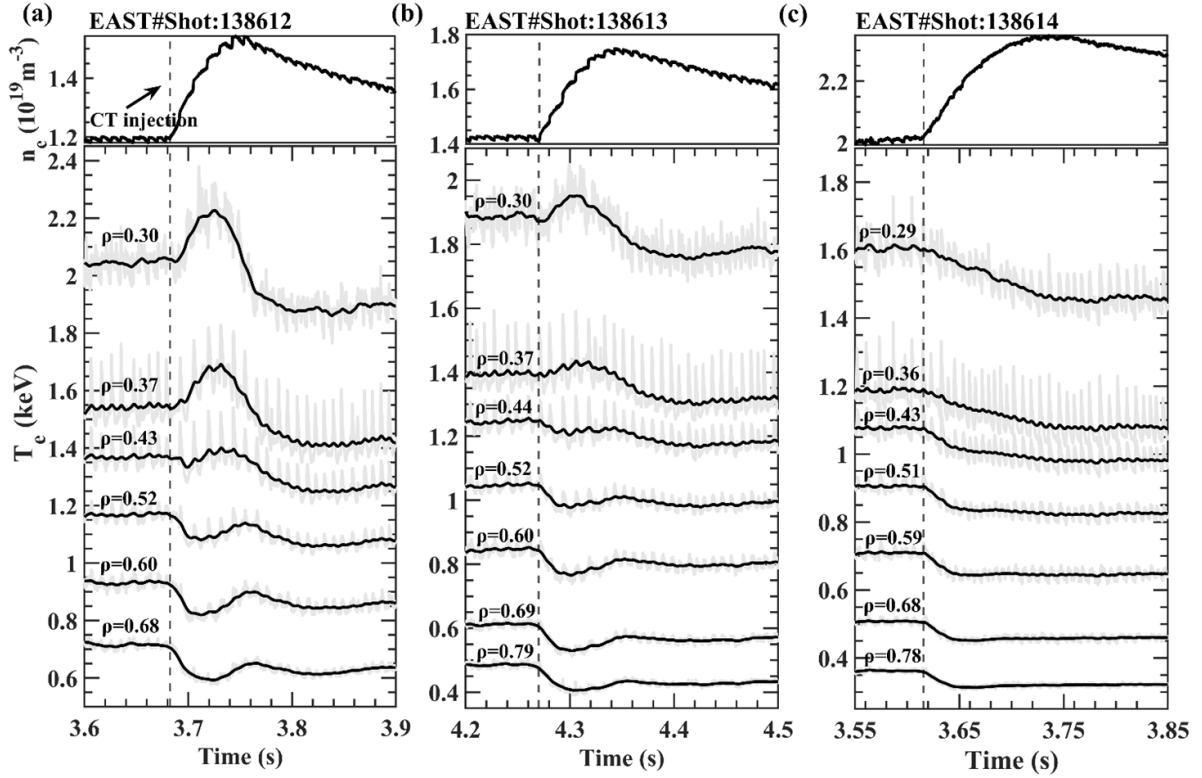


Figure 3. The waveforms of electron temperature after CT injection under different three baseline levels of the line-averaged electron densities (from POINT interferometer): (a) low-density ($\bar{n}_e = 1.2 \times 10^{19} \text{m}^{-3}$), (b) medium-density plasma ($\bar{n}_e = 1.4 \times 10^{19} \text{m}^{-3}$), (c) high-density plasma ($\bar{n}_e = 2.0 \times 10^{19} \text{m}^{-3}$).

phenomenon resulting from cold pulse injection is strongly density-dependent.

Figure 4 show the electron density and temperature profiles at the time of CT injection and shortly following CT injection. As shown in figure 4(a), compared with the electron density profile at the time of CT injection ($t_0 = 3682$ ms, black curve), the electron density profile measured 4 ms after CT injection (pink curve) increased at almost all radial locations. At 10 ms, 20 ms and 30 ms after CT injection (purple, green and blue curves), a pronounced increase in the electron density was observed in the core region. This could be attributed to the continuous inward particle diffusion induced by the CT injection. The dynamic response of the density gradient was further quantified in figure 4(b), which shows the evolution of the normalized electron density gradient (a/L_{n_e}). Immediately following the CT injection, a significant reduction in a/L_{n_e} is observed in the core region ($\rho < 0.4$), indicating a rapid flattening of the density profile. This localized flattening might be the primary driver for the subsequent stabilization of microturbulence. Figures 4(c) and (d) represent the evolution of electron temperature profiles and the corresponding temperature changes ($\Delta T_e = T_e(t) - T_e(t_0)$). As demonstrated in figure 4(d), a clear temperature reversal phenomenon is captured. The edge temperature ($\rho > 0.45$) decreases due to the cold pulse, the core temperature ($\rho < 0.45$) exhibited a significant increase. A temperature reversal is observed at $\rho \sim 0.4$ – 0.5 , and the region of temperature increase aligns closely with the region with density flattening region. It is also interesting

to point out that the density profile momentarily peaked around $\rho \sim 0.3$, which indicates that the particles in the CT had already been deposited to the core plasma, at a location deeper than where the reversal of the electron temperature occurred.

Figure 5 shows the evolution of the relative density variations and the increase in electron temperature at different radial positions at during and after CT injection in the EAST discharge #138612. As shown in figure 5(a), the relative electron density variation, defined by $(n_e - n_{e0})/n_{e0}$ (where n_{e0} is the electron density at the time of CT injection), reveals that the electron density propagates from the plasma edge to the core on a very fast timescale following CT injection. At the plasma core ($\rho \sim 0.37$) the density pulse was observed to arrive just 2 ms after CT injection. The rise in the core electron temperature, which began around 4 ms after CT injection (figure 5(b)), lagged behind the initial increase in core density. This is because the CT can rapidly deposit particles into the plasma core, directly altering the electron density at that position.

To further clarify the relationship between the changes in core temperature and core density, the evolution of core electron temperature (T_e) is compared with the normalized electron density gradient scale (a/L_{n_e}) and the normalized electron temperature gradient scale (a/L_{T_e}) as shown in figure 6. During the interval marked by the red box (Δt_1 , approximately 3682 ms to 3686 ms), the normalized core electron temperature varied only gradually, which corresponded to a similarly small variation in the normalized electron temperature

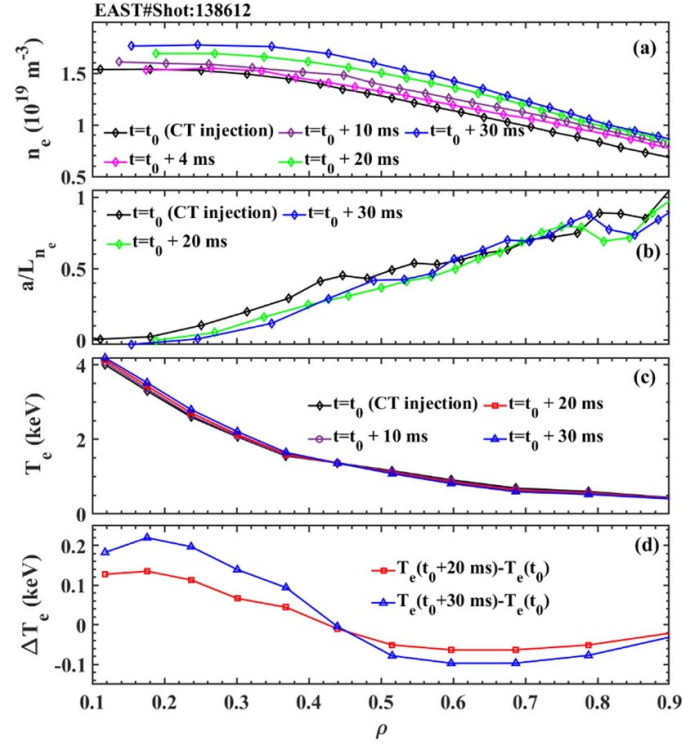


Figure 4. Radial profiles of plasma parameters before and after CT injection in EAST: (a) electron density, (b) normalized electron density gradient length, (c) electron temperature, and (d) the change in electron temperature relative to the pre-injection baseline.

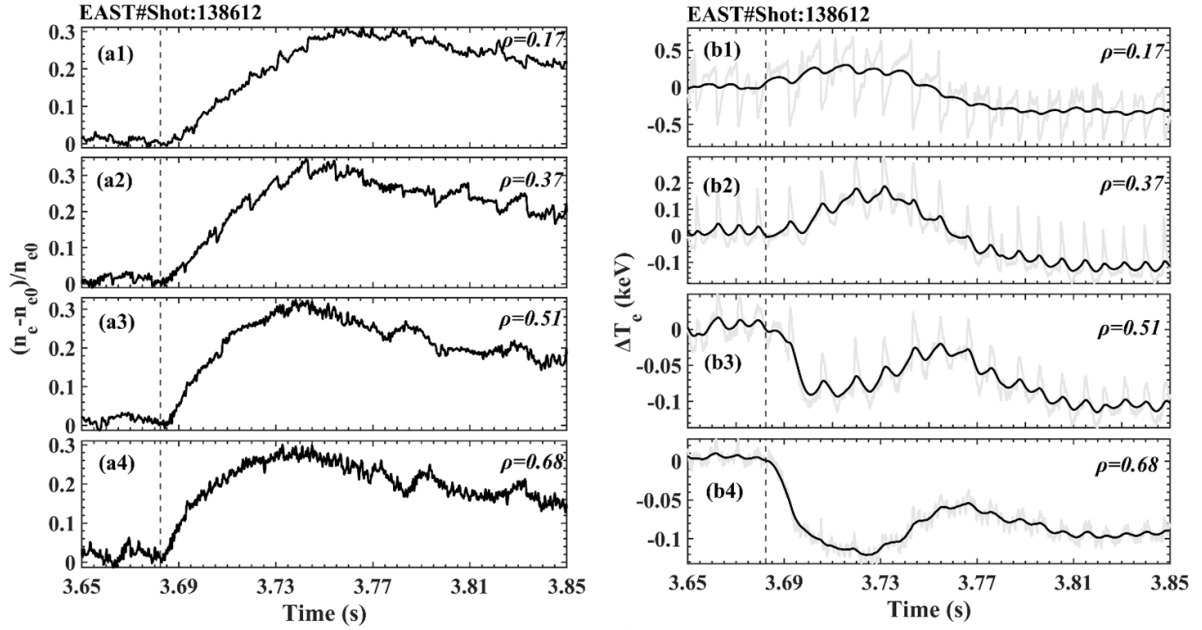


Figure 5. Evolution of relative variations in electron temperature and electron density at different radial positions around the time of CT injection.

gradient. In contrast, the normalized electron density gradient around the same core region exhibited a sharp change. This suggests that the CT injection first altered the normalized core electron density gradient, which then caused the core profile to flatten rapidly. Upon entering the time interval marked by the

cyan box (Δt_2 , approximately from 3686 ms to 3727 ms), the core electron temperature rose to a peak. Correspondingly, the normalized temperature gradient first decreases continuously and then gradually recovered to near the horizontal line.

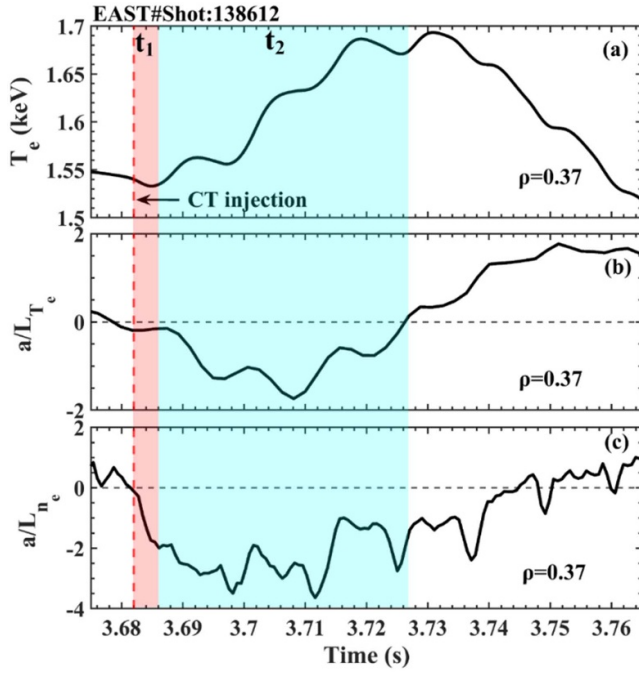


Figure 6. The evolution of (a) core temperature T_e , (b) the normalized reversal gradient scale of temperature gradient a/L_{T_e} and (c) the normalized reversal gradient scale of the density a/L_{n_e} after CT injection.

3.3. Experimental observation of QC-TEM and micro-instabilities simulations

The simulation results of Alcator C-Mod indicate that the stabilization of the turbulent TEM caused by the sudden flattening of the electron density profile is responsible for the increase in the core temperature [16]. Although it is challenging to distinguish between ITG modes and TEM modes experimentally due to the overlap of their characteristic scale ranges, a large body of existing experimental evidence indicates that the evolutionary behavior of the relevant turbulence can be determined by identifying the direct spectral characteristics of fluctuations induced by coherent mode instabilities [36]. As shown in figure 7, the spectral measurements of the plasma core obtained via the DBS system, there are two distinct symmetric peaks (70–110 kHz) at the core region. These symmetric peaks suggest the presence of quasi-coherent trapped electron (QC-TEM) modes, which have been observed in similar plasma conditions on various devices [37–40]. To rigorously confirm this identification, integrated kinetic simulations were performed. A comparison of the spectral features before and after CT injection shows that the intensity of QC-TEM decreased after CT injection in shot 138613 with a low baseline level of the averaged density, suggesting that the TEM stabilization and might providing direct evidence of the turbulence evolution. For shot 138614 with higher line-averaged density, the QC-TEM was both initially weak and largely unresponsive to CT injection; correspondingly, no increase in core temperature was observed.

To determine whether the core temperature increase resulted from TEM turbulence stabilization, a numerical study was conducted on the plasma turbulence characteristics during this CT-induced heating period. This was achieved by leveraging the TGLF model in combination with previous experimental results [41–43]. Figure 8(a) shows the evolution of linear growth rate and the real frequencies before and after CT injection. Before CT injection, the core plasma ($\rho = 0.18$) was dominated by the low- k TEM turbulence. This is consistent with the expectation that, under ECRH heating, the core plasma turbulence in EAST is dominated by TEM. After CT injection, the growth rate of the low- k mode decreased significantly, which might have been the cause of the increase in the core temperature. This was precisely caused by the rapid flattening of the core electron density profile resulting from the CT injection. As illustrated in figure 8(b), the real frequency of the most unstable modes remained in the electron diamagnetic direction (positive values) both before and after CT injection at $\rho = 0.18$. In many transient transport studies [16, 17], a sign reversal of the real frequency is considered a classic signature of a transition from TEM-dominated to ITG-dominated turbulence. However, no such transition was observed here. This persistence of TEM dominance is likely attributed to the strong ECRH, as previous simulation studies [18] have consistently shown that under dominant electron heating conditions, TEM can remain the primary instability even during significant density profile modifications.

As depicted in figures 8(c) and (d), which present the growth rate and real frequencies at the plasma position of $\rho = 0.62$, the growth rate exhibited minimal change following CT injection. Simulation results indicate that the low- k TEM turbulence at the core of the plasma is stabilized by CT injection, while the turbulence in the region outside the temperature reversal location remains unaffected. This simulation result was consistent with the observed suppression of QC-TEM observed in experiments. Both experimental and numerical evidence suggest that in low-density discharges, CT injection induces rapid changes in core plasma density, leading to TEM stabilization and, consequently, an increase in core electron temperature.

4. Discussion and summary

Based on the characteristics of CT injection, this study conducted experiments on core temperature reversal caused by cold pulse injection on EAST, expanding the understanding of ‘non-local’ heat transport phenomena. These phenomena, such as core temperature reversal and near instantaneous thermal responses, have traditionally been labeled as ‘non-local’ in a phenomenological context [2, 16], and not in a strict mathematical sense, as it is usual e.g. in fractional kinetics [44]. Indeed, our results demonstrate that the aforementioned phenomena are deeply rooted in local turbulence dynamics. In this experiment, core temperature reversal in low density, and lack of reversal at high density were observed. The measurement using the reflectometer revealed an increase in the

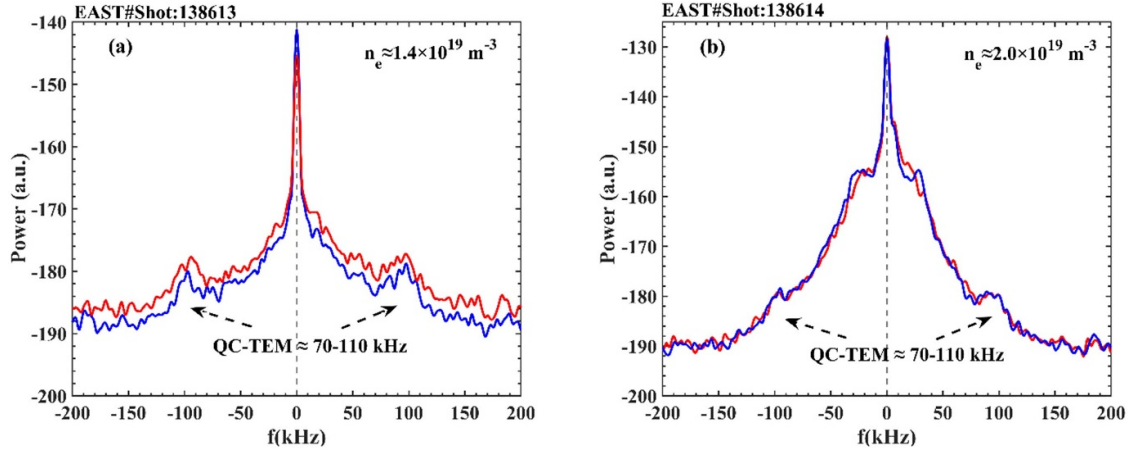


Figure 7. Core plasma fluctuation spectra the signals DBS during the CT fueled EAST discharge #138613 and #138614. The red line represents the spectra before CT injection, and the blue line represents that after CT injection.

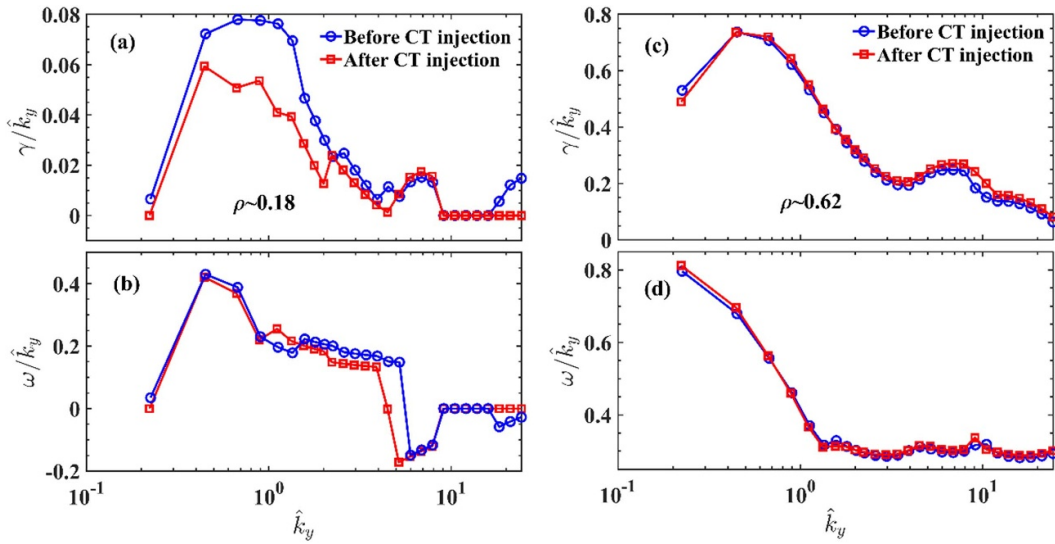


Figure 8. Obtained through TGLF the linear growth rate (a) and (c), mode frequency (b) and (d) at $\rho = 0.18$ and $\rho = 0.62$. Blue line and the red line represent the simulated growth rates and frequencies before and after CT injection, respectively.

core electron density at 2 ms after CT injection, which caused the core electron density profile to flatten rapidly. Detailed kinetic analysis confirms a clear causal sequence. The CT injection first triggered a rapid flattening of the normalized electron density gradient, which subsequently stabilized the TEM turbulence and drove the rise in core electron temperature. The symmetrical spectral peaks captured by the DBS system in the core were identified as QC-TEM. This physical picture was rigorously supported by TGLF simulations, which confirmed that low- k TEM turbulence in the core was stabilized by the flattened density profile under low-density conditions.

Our study demonstrates that the ‘non-local’ signatures induced by CT can be self-consistently explained by the local suppression of TEM turbulence following rapid density profile flattening. By employing CT injection as a novel main plasma particle fueling method, this work established

a robust causal link between deep fueling and localized transport improvement. The direct observation of QC-TEM stabilization serves as direct evidence that the core thermal response is governed by local gradient changes. This study not only resolves the underlying mechanism of temperature reversal on EAST but also provided vital experimental insights for active transport control in magnetically confined fusion plasma.

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References

- [1] Rodriguez-Fernandez P., Rice J.E., Cao N.M., Creely A.J., Howard N.T., Hubbard A.E., Irby J.H. and White A.E. 2017 *Nucl. Fusion* **57** 074001
- [2] Gentle K.W. et al 1995 *Phys. Rev. Lett.* **74** 3620–3
- [3] Kissick M.W., Callen J.D., Fredrickson E.D., Janos A.C. and Taylor G. 1996 *Nucl. Fusion* **36** 1691–701
- [4] Zou X.L. et al 2000 *Plasma Phys. Control. Fusion* **42** 1067–76
- [5] Ryter F. et al 2000 *Nucl. Fusion* **40** 1917–32
- [6] Mantica P. et al 2002 *Plasma Phys. Control. Fusion* **44** 2185–215
- [7] Sun H.J. et al 2010 *Plasma Phys. Control. Fusion* **52** 045003
- [8] Liu Y. et al 2019 *Nucl. Fusion* **59** 044005
- [9] Rice J.E. et al 2013 *Nucl. Fusion* **53** 033004
- [10] Shi Y.J. et al 2017 *Nucl. Fusion* **57** 066040
- [11] Shi Y.J. et al 2018 *Nucl. Fusion* **58** 044002
- [12] Inagaki S. et al 2006 *Plasma Phys. Control. Fusion* **48** A251–7
- [13] Mantica P., Galli P., Gorini G., Hogeweij G.M.D., de Kloe J. and Lopes Cardozo N.J. (RTP Team) 1999 *Phys. Rev. Lett.* **82** 5048–51
- [14] Del-castillo-negrete D., Mantica P., Naulin V. and Rasmussen J.J. 2008 *Nucl. Fusion* **48** 075009
- [15] Hariri F., Naulin V., Juul Rasmussen J., Xu G.S. and Yan N. 2016 *Phys. Plasmas* **23** 052512
- [16] Rodriguez-Fernandez P. et al 2018 *Phys. Rev. Lett.* **120** 075001
- [17] Rodriguez-Fernandez P. et al 2019 *Phys. Plasmas* **26** 062503
- [18] Angioni C., Fable E., Ryter F., Rodriguez-Fernandez P. and Pütterich T. 2019 *Nucl. Fusion* **59** 106007
- [19] Ye Y. et al 2025 *Nucl. Fusion* **65** 084002
- [20] Raman R., Thomas J.C., Hwang D.Q., Conway G.D., Martin F., Hirose A., Gierszewski P. and Décoste R. 1993 *Fusion Technol.* **24** 239–50
- [21] Raman R. et al 1994 *Phys. Rev. Lett.* **73** 3101
- [22] Raman R. et al 1997 *Nucl. Fusion* **37** 967
- [23] Nagata M. et al 2005 *Nucl. Fusion* **45** 1056
- [24] Li J. et al 2013 *Nat. Phys.* **9** 817–21
- [25] Li J.G. et al 2021 *Engineering* **7** 1523–8
- [26] Liu X., Zhao H.L., Liu Y., Li E.Z., Han X., Domier C.W., Luhmann N.C., Ti A., Hu L.Q. and Zhang X.D. 2014 *Rev. Sci. Instrum.* **85** 093508
- [27] Qu H. et al 2015 *Plasma Sci. Technol.* **17** 985
- [28] Liu H.Q. et al 2016 *Rev. Sci. Instrum.* **87** 11D903
- [29] Hu J.Q. et al 2017 *Rev. Sci. Instrum.* **88** 073504
- [30] Liu S. et al 2023 *Rev. Sci. Instrum.* **94** 123507
- [31] Feng X. et al 2019 *Rev. Sci. Instrum.* **90** 024704
- [32] Tan M.S. et al 2024 *Fusion Eng. Des.* **205** 114559
- [33] Zhao Z.H. et al 2025 *Plasma Sci. Technol.* **27** 065601
- [34] Parks P.B. 1988 *Phys. Rev. Lett.* **61** 1364
- [35] Perkins J., Ho S.K. and Hammer J.H. 1988 *Nucl. Fusion* **28** 1365
- [36] Arnichand H. et al 2015 *Nucl. Fusion* **55** 093021
- [37] Arnichand H. et al 2016 *Plasma Phys. Control. Fusion* **58** 014037
- [38] Citrin J., Arnichand H., Bernardo J., Bourdelle C., Garbet X., Jenko F., Hacquin S., Pueschel M.J. and Sabot R. 2017 *Plasma Phys. Control. Fusion* **59** 064010
- [39] Zhong W.L. et al 2016 *Phys. Plasmas* **23** 060702
- [40] Arnichand H. et al 2014 *Nucl. Fusion* **54** 123017
- [41] Staebler G.M., Kinsey J.E. and Waltz R.E. 2007 *Phys. Plasmas* **14** 055909
- [42] Kinsey J.E., Staebler G.M. and Waltz R.E. 2008 *Phys. Plasmas* **15** 055908
- [43] Nan K.B. et al 2024 *Plasma Phys. Control. Fusion* **66** 115015
- [44] Zaslavsky G.M. 2005 *Hamiltonian Chaos and Fractional Dynamics* vol 16 (Oxford University Press) pp 245–59